BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(MID SEMESTER EXAMINATION SP/2024)

CLASS: B.TECH | **SEMESTER : II**
BRANCH: CSE/AIML/ECE/EEE | **SESSION : SP/2024**

SUBJECT: EE101 BASIC OF ELECTRICAL ENGINEERING

TIME: 02 Hours | **FULL MARKS: 25**

INSTRUCTIONS:
1. The question paper contains 5 questions each of 5 marks and total 25 marks.
2. Attempt all questions.
3. The missing data, if any, may be assumed suitably.
***4*. Tables/Data handbook/Graph paper etc., if applicable, will be supplied to the candidates**

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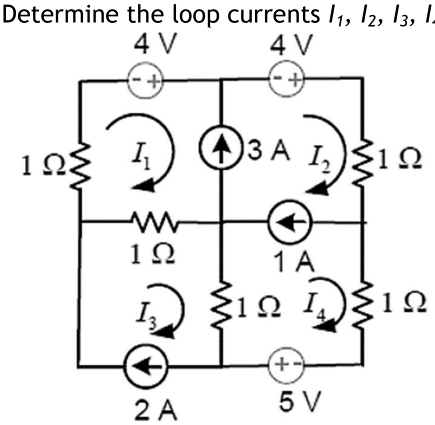
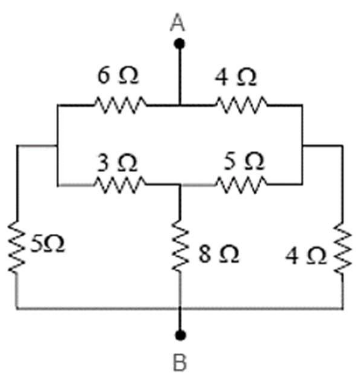
Q.1(a) Define the terms: (i) Independent Source (ii) Dependent Source [2] 1 1

Q.1(b) Determine the loop currents I_1, I_2, I_3, I_4 for the network shown in the figure below: [3] 1 3



Q.2(a) Explain the concept of a Super node and state the limitations associated in this analysis? [2] 1 2

Q.2(b) Using the Star/Delta conversion technique, determine the current flowing between the terminals A and B when a 5V dc source is connected between them? [3] 1 3



Q.3(a) Define the concept of a power triangle with a neat phasor diagram? [2] 2 1

Q.3(b) Four emfs, $e_1 = 100\sin(wt)$, $e_2 = 80\sin(wt-\pi/3)$, $e_3 = 90\sin(wt+\pi/4)$ and $e_4 = 120\cos(wt)$ are induced in four coils connected in series. Assuming that the current ($i = I\sin(wt)$) flowing in the circuit lags e_4, calculate the resultant emf and its phase difference with e_1 and e_2. Draw a neat phasor diagram? [3] 2 3

Q.4(a) Define the terms: (i) Average Value (ii) RMS Value (iii) Form Factor (iv) Phase Difference [2] 2 1

Q.4(b) In a series RLC circuit, an AC voltage of $120e^{+j0}$ V is applied at a frequency of 400 rad/sec. The input current leads the voltage by 63.5 degrees. Calculate the value of 'R' if L = 0.025H and C = 50μF. What are the voltage drops across L and C? Draw a neat phasor diagram? [3] 2 3

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Q.5(a)	Explain the concept of a series resonance through the use of impedance-frequency curve. Plot the Impedance frequency curve also?	[2]	2	2
Q.5(b)	A series resonant RLC circuit has a resonant frequency of 80K rad/sec and a quality factor of 8. Calculate the bandwidth, the upper cut-off and lower cut-off frequencies?	[3]	2	3

:::::23/02/2024 E:::::